

DEVELOPER BLUEPRINT DOCUMENT

PulseVision: Multi-Dataset Ensemble Framework for PCG Heart Sound Classification

Implementation Plan, Dataset Mapping, Proposed Algorithm, Comparative Experiments, Tables, Graphs, Equations, Q1 Paper Results, Thesis Deliverables, Dashboard and Developer Tasks

Project Type	PhD-oriented academic research prototype
Implementation Nature	Python-based software system; no hardware integration; no paid API dependency
Proposed Framework Name	PV-MEPCG: PulseVision Multi-Dataset Ensemble PCG Framework
Primary Model Stack	SVM-RBF + Random Forest + Gradient Boosting + optimized soft voting
Core Inputs	PCG WAV recordings from PhysioNet 2016, PASCAL Heart Sounds and CirCor DigiScope 2022
Key Outputs	Normal/Abnormal prediction, multiclass heart-sound type, murmur status, confidence, feature importance, robustness analytics and exportable report
Prepared For	Developer implementation, thesis result generation, IEEE/Q1 paper preparation and client demonstration

Important Scope Boundary

The system must be presented as an academic screening and decision-support prototype. It must not be claimed as a certified diagnostic tool, a hospital product, a medical device or a replacement for clinical judgement.

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1. Final Locked Objectives

Non-Negotiable Instruction

The following objectives are locked. Do not change, shorten, paraphrase or reword them in implementation documents, reports, thesis chapters, paper drafts or client-facing material.

Objective No.	Locked Objective Wording
Objective 1	To introduce a new approach for the classification of normal and abnormal heart sound recordings using a machine learning algorithm. An appropriate algorithm for auto diagnosis systems of heart diseases, capable of distinguishing most known pathological states, must be developed.
Objective 2	To review recently published preprocessing, feature extraction, and classification techniques, as well as the state-of-the-art in phonocardiogram (PCG) signal analysis.
Objective 3	To utilise ensemble machine learning algorithms that combine the predictions of several learning models into a single “ensemble” model to improve overall performance.
Objective 4	To understand heart sound classification techniques that might find use in medical diagnostic systems; the objective is to accurately classify normal and abnormal heart sounds from single, short, and potentially noisy recordings.
Objective 5	Apply a search algorithm to improve the performance of the diagnostic system in terms of accuracy, complexity, and the range of distinguishable heart sounds.
Objective 6	To introduce a greater number of classes (types of heart sounds) to test the final system further, making it capable of handling more tests on a larger number of samples.

2. Dataset Inventory and Role

Dataset / Subset	Role	Prediction Task	Key Characteristics	Status
PhysioNet/CinC 2016	Primary binary PCG benchmark	Normal vs Abnormal	~3,126 recordings; WAV; 2,000 Hz; real-world noise	Mandatory
PASCAL Heart Sounds Dataset A	Primary multiclass acoustic dataset	Normal, Murmur, Extra Heart Sound, Artifact	Small but explicitly labelled multiclass set	Mandatory for Objective 6
PASCAL Heart Sounds Dataset B	Secondary multiclass acoustic dataset	Normal, Murmur, Extrasystole	Useful for a second multiclass experiment and noisy cases	Mandatory for Objective 6
CirCor DigiScope 2022	Large external and multi-location validation	Murmur Present/Absent/Unknown; Normal/Abnormal outcome	5,272 recordings from 1,568 subjects; multi-location metadata	Strongly recommended

Official download sources:

- [PhysioNet/CinC 2016](#)
- [PASCAL Heart Sounds Challenge](#)
- [CirCor DigiScope v1.0.3](#)

3. Objective-wise Dataset and Implementation Mapping

Objective	Dataset(s)	Implementation Mapping	Required Evidence
Objective 1	PhysioNet 2016 + CirCor 2022	Develop the primary binary ensemble on PhysioNet and verify generalization using CirCor outcome/murmur labels.	Binary metrics, ROC/PR curves, confusion matrix, external validation.
Objective 2	All three dataset families	Compare preprocessing, feature families and classification methods across different noise, duration, label and population conditions.	Literature review table, preprocessing ablation and feature-family ablation.
Objective 3	PhysioNet 2016 + CirCor 2022	Train SVM, RF and GB individually; combine them through optimized soft voting; confirm ensemble behavior externally.	Individual-vs-ensemble comparison, calibration and cross-dataset generalization.
Objective 4	PhysioNet 2016 + PASCAL + CirCor	Evaluate short, noisy, low-confidence and multi-location recordings.	Duration-wise, noise-wise and dataset-wise robustness tables.
Objective 5	PhysioNet 2016 for search; PASCAL/CirCor for confirmation	Use search methods for feature selection, hyperparameter tuning and ensemble-weight optimization.	Baseline-vs-optimized results, selected features, inference time and complexity trade-off.
Objective 6	PASCAL A/B + CirCor	Implement genuine multiclass acoustic classification and large-sample murmur/outcome evaluation.	Multiclass confusion matrices, macro-F1, per-class recall and large-sample validation.

4. Proposed Algorithm and Novelty

Final Proposed Framework Name

PV-MEPCG: PulseVision Multi-Dataset Ensemble PCG Classification Framework

PV-MEPCG is not a single plain classifier. It is an integrated framework combining standardized PCG preprocessing, unified multi-domain feature engineering, search-based feature and hyperparameter optimization, heterogeneous ensemble learning, binary and multiclass prediction, external validation, robustness analysis and explainable reporting.

Novelty Component	Description
Unified multi-domain feature stack	Approximately 138 features from time, frequency, MFCC, chroma, DWT and envelope domains are represented as one fixed-length vector.
Search-optimized feature subset	A search module selects a compact, high-value subset to improve performance and reduce complexity.
Optimized heterogeneous ensemble	SVM-RBF, Random Forest and Gradient Boosting are combined through probability-based soft voting with tunable ensemble weights.
Binary + multiclass integration	The same core pipeline supports Normal/Abnormal classification, multiclass acoustic-event classification and murmur/outcome analysis.
Multi-dataset validation	The framework is evaluated on PhysioNet, PASCAL and CirCor rather than one dataset only.
Noise and duration robustness	Performance is explicitly analysed for noisy, short-duration and low-confidence recordings.
Results-aware transparency	Prediction, confidence, feature importance, confusion matrices, robustness graphs and downloadable reports are exposed through the dashboard.
Performance-complexity trade-off	The project evaluates accuracy, macro-F1, feature count, model size and inference time together.

5. End-to-End System Architecture

Step	Module	Developer Action
Step 1	PCG input and metadata loading	Load WAV file, dataset source, label, subject ID, recording location and sampling rate.
Step 2	Signal harmonization	Convert to mono and resample to 2,000 Hz.
Step 3	Signal preprocessing	Apply 20-400 Hz 4th-order Butterworth bandpass and z-score normalization.
Step 4	Signal quality analysis	Calculate duration, clipping/noise indicators and low-confidence flags.
Step 5	Feature extraction	Generate approximately 138 time, frequency, MFCC, chroma, DWT and envelope features.
Step 6	Feature scaling	Fit scaler on training fold only and transform validation/test data.
Step 7	Search optimization	Optimize feature subset, model

		hyperparameters and ensemble weights.
Step 8	Individual model training	Train SVM-RBF, Random Forest and Gradient Boosting.
Step 9	Soft-voting ensemble	Average or weight calibrated class probabilities.
Step 10	Task routing	Produce binary, multiclass or murmur/outcome prediction depending on dataset and experiment.
Step 11	Explainability and analytics	Generate feature importance, confidence, error analysis and dataset-wise robustness.
Step 12	Dashboard and report	Display outputs and export sample-level and experiment-level reports.

6. Data Harmonization and Leakage-Control Protocol

- Create one master metadata table with dataset_source, subject_id, recording_id, file_path, original_sampling_rate, duration_sec, recording_location, binary_label, multiclass_label, murmur_label and split_group.
- Convert all recordings to mono WAV and resample to a common 2,000 Hz rate.
- Do not mix incompatible class labels into a single target. Keep binary, multiclass and murmur/outcome tasks separate.
- Fit filtering-dependent transforms, scaling and feature selection inside each training fold.
- Use GroupKFold or StratifiedGroupKFold when subject IDs are available.
- Never place recordings from the same subject in both training and testing.
- Retain dataset source for domain-shift analysis.
- Detect duplicate and near-duplicate recordings before splitting.
- Use class weights and macro metrics for imbalanced experiments.

7. Algorithm and Search-Method Comparison Plan

Algorithm	Category	Priority	Purpose
Logistic Regression	Linear baseline	Mandatory	Simple, interpretable benchmark
K-Nearest Neighbours	Distance-based baseline	Optional	Shows effect of local feature similarity
SVM-RBF	Primary model 1	Mandatory	Strong nonlinear boundary for engineered features
Random Forest	Primary model 2	Mandatory	Robust tree ensemble and feature importance
Gradient Boosting	Primary model 3	Mandatory	Sequential error correction
Soft Voting Ensemble	Proposed final model	Mandatory	Combines calibrated probabilities
XGBoost or LightGBM	Strong external baseline	Recommended	Useful publication-grade comparison if environment permits
1D-CNN	Deep-learning baseline	Optional	Compares handcrafted features with raw-signal learning
Search Method	Type	Priority	Use

Grid Search	Deterministic exhaustive search	Baseline tuning	Use on small parameter spaces
RandomizedSearchCV	Efficient random exploration	Mandatory initial search	Fast and practical
Bayesian Optimization	Sequential model-based optimization	Recommended final search	Efficient for expensive search spaces
Genetic Algorithm	Evolutionary feature/hyperparameter search	Optional high-novelty module	Useful for feature subset + ensemble weights
Particle Swarm Optimization	Population-based optimization	Optional alternative	Useful for multi-objective optimization

8. Experimental Tracks and Ablation Plan

Track	Dataset	Task	Priority	Required Work
Track A	PhysioNet 2016	Binary Normal vs Abnormal	Mandatory	Primary benchmark; all models + ensemble + search
Track B	PASCAL Dataset A	4-class acoustic classification	Mandatory	Normal/Murmur/Extra Heart Sound/Artifact
Track C	PASCAL Dataset B	3-class acoustic classification	Mandatory	Normal/Murmur/Extrasystole
Track D	CirCor 2022	Murmur and outcome classification	Strongly recommended	Subject-wise, multi-location external validation
Track E	Cross-dataset	Train PhysioNet, test compatible CirCor binary outcome	Recommended	Generalization evidence
Track F	Ablation	Feature families and optimization modules	Mandatory for publication	Novelty proof
Ablation ID		Configuration	Purpose	
A1		Time-domain only	Quantify temporal contribution	
A2		Frequency-domain only	Quantify spectral contribution	
A3		MFCC only	Quantify cepstral contribution	
A4		Wavelet only	Quantify non-stationary contribution	
A5		Time + Frequency	Classical baseline	
A6		MFCC + Wavelet	Acoustic and multiresolution fusion	
A7		All 138 features	Full proposed representation	
A8		Optimized feature subset	Performance-complexity contribution	
A9		Best individual model	Compare against ensemble	
A10		Equal-weight ensemble vs optimized-weight ensemble	Prove search contribution	

9. Tables Required for Thesis and Q1 Paper

No.	Table Name	Content Required	Use
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Table 1	Dataset Inventory	Dataset name, source, subjects, recordings, classes, sampling rate, role	Thesis + Paper
Table 2	Class Distribution	Class-wise sample counts and imbalance ratio	Thesis + Paper
Table 3	Preprocessing Settings	Resampling, filter, normalization and split protocol	Thesis
Table 4	Feature Inventory	All feature families and counts totaling approximately 138	Thesis + Paper
Table 5	Model Configuration	Hyperparameters of all baseline and ensemble models	Thesis + Paper
Table 6	Search Space	Search variables, ranges and final selected values	Thesis + Paper
Table 7	PhysioNet Binary Comparison	Accuracy, precision, recall, specificity, F1, ROC-AUC, PR-AUC	Thesis + Paper
Table 8	PASCAL Dataset A Results	Macro-F1, weighted-F1, balanced accuracy and per-class recall	Thesis + Paper
Table 9	PASCAL Dataset B Results	Macro-F1, weighted-F1 and class-wise metrics	Thesis + Paper
Table 10	CirCor External Results	Murmur/outcome performance and subject-wise validation	Q1 high-value
Table 11	Cross-Dataset Generalization	Train dataset, test dataset, performance drop	Q1 high-value
Table 12	Feature Ablation	Feature family vs performance	Q1 high-value
Table 13	Optimization Ablation	Baseline vs tuned model vs optimized ensemble	Q1 high-value
Table 14	Complexity Analysis	Feature count, training time, inference time, model size	Thesis + Paper
Table 15	Noise Robustness	Clean/noisy/low-confidence performance	Q1 high-value
Table 16	Duration Robustness	Short/medium/long recording performance	Q1 high-value
Table 17	Calibration and Confidence	ECE/Brier score or confidence bins	Paper
Table 18	Failure Analysis	False positive/false negative categories	Thesis
Table 19	Objective Achievement	Objective, module, dataset, proof and status	Thesis
Table 20	Final System Outputs	Sample file, prediction, confidence, task, runtime	Thesis

10. Graphs, Charts and Visual Outputs Required

Graph ID	Graph / Chart	Purpose
G1	Dataset class-distribution bar chart	Show class imbalance
G2	Waveform before/after filtering	Show preprocessing effect
G3	Spectrogram comparison	Visualize normal vs abnormal frequency patterns
G4	Feature-family count chart	Show 138-feature composition

G5	Model comparison bar chart	Compare individual models and ensemble
G6	ROC curve	Binary discrimination
G7	Precision-recall curve	Imbalanced-data evaluation
G8	Binary confusion matrix	Class-wise errors
G9	Multiclass confusion matrix A	PASCAL Dataset A
G10	Multiclass confusion matrix B	PASCAL Dataset B
G11	Feature-importance plot	Explain influential features
G12	Search convergence plot	Show optimization progress
G13	Accuracy/F1 vs feature count	Show performance-complexity trade-off
G14	Inference time vs performance	Deployment-oriented comparison
G15	Dataset-wise generalization chart	Show domain shift
G16	Noise-level robustness chart	Performance by noise condition
G17	Duration-wise performance chart	Short, medium and long recordings
G18	Confidence histogram	Prediction certainty
G19	Calibration curve	Probability reliability
G20	Error-analysis chart	False positive/negative categories

11. Equations and Optimization Formulas

Formula	Equation	Use
Butterworth response	$ H(j\omega) ^2 = 1 / [1 + (\omega/\omega_c)^{2n}]$	Preprocessing
Z-score	$z = (x - \mu) / \sigma$	Signal normalization
Energy	$E = \sum x[n]^2$	Time-domain feature
RMS	$RMS = \sqrt{[(1/N)\sum x[n]^2]}$	Effective amplitude
Spectral centroid	$C = \sum f_k X_k / \sum X_k$	Spectral centre of mass
Spectral entropy	$H_s = -\sum p_k \log(p_k)$	Spectral irregularity
Soft voting	$P_{final}(c) = \sum w_m P_m(c) / \sum w_m$	Weighted ensemble probability
Prediction	$\hat{y} = \operatorname{argmax}_c P_{final}(c)$	Final class
Accuracy	$(TP+TN)/(TP+TN+FP+FN)$	Overall correctness
Sensitivity	$TP/(TP+FN)$	Abnormal-case detection
Specificity	$TN/(TN+FP)$	Normal-case detection
F1-score	$2PR/(P+R)$	Precision-recall balance
Balanced accuracy	$(Sensitivity+Specificity)/2$	Imbalance-aware binary metric
Macro-F1	Mean of class-wise F1	Multiclass evaluation
Multi-objective score	$J = \alpha(1-MacroF1) + \beta(SelectedFeatures/138) + \gamma(NormalizedInferenceTime)$	Optimization objective

12. Binary Classification Module

- Primary dataset: PhysioNet/CinC 2016.

- Labels: Normal=0 and Abnormal=1 internally; preserve original labels in metadata.
- Mandatory models: Logistic Regression, SVM-RBF, Random Forest, Gradient Boosting and soft-voting ensemble.
- Validation: grouped or stratified 5-fold cross-validation; use subject grouping where available.
- Primary metrics: sensitivity, specificity, F1, balanced accuracy, ROC-AUC, PR-AUC and confusion matrix.
- Final model selection rule: prioritize sensitivity and balanced accuracy, not accuracy alone.
- Output: predicted class, confidence, model probabilities, inference time and low-confidence flag.

13. Multiclass Classification Module

- PASCAL Dataset A task: Normal, Murmur, Extra Heart Sound and Artifact.
- PASCAL Dataset B task: Normal, Murmur and Extrasystole.
- Use the same preprocessing and feature-extraction pipeline.
- Use one-vs-rest SVM or native multiclass models as appropriate.
- Report macro-F1, weighted-F1, balanced accuracy, per-class precision/recall, one-vs-rest AUC and confusion matrix.
- Use repeated validation because PASCAL is relatively small.
- Do not merge Dataset A and B labels blindly; report them as separate experiments.

14. External Validation and Robustness Module

- Use CirCor DigiScope 2022 for subject-wise murmur and outcome analysis.
- Keep all recordings of the same subject in one split.
- Evaluate both recording-level and subject-level aggregation.
- Analyse performance by auscultation location where metadata permits.
- Perform cross-dataset testing only for compatible labels.
- Report domain-shift degradation honestly.
- Analyse short-duration, noisy and low-confidence cases separately.
- Generate a failure-case report for false positives and false negatives.

15. Dashboard Blueprint

Dashboard Page	Required Features
Home	Project objective, pipeline, disclaimer and dataset summary
Dataset Overview	Dataset names, classes, sample counts, recording duration and split
Signal Preprocessing	Waveform, filtering, normalization and signal-quality indicators
Feature Extraction	138-feature summary, family counts and selected subset
Model Comparison	Individual-model and ensemble metrics
Search Optimization	Search method, convergence, selected parameters and selected features
Binary Prediction	Upload WAV and show Normal/Abnormal result with confidence
Multiclass Prediction	Show PASCAL class output and class probabilities
Murmur/Outcome Analysis	CirCor recording-level and subject-level output
Robustness Analytics	Noise, duration, dataset and location-wise results

Explainability	Feature importance and sample-level explanation
Reports	Export sample prediction, experiment results and objective-coverage report

16. Q1 / IEEE Paper Result Requirements

Paper Positioning

Do not position the paper as a simple Normal-vs-Abnormal classifier. Position it as a multi-dataset, search-optimized, explainable ensemble framework for binary and multiclass PCG classification with external validation and performance-complexity analysis.

- Unified approximately 138-feature multi-domain PCG representation.
- Search-optimized compact feature subset and model configuration.
- Optimized probability-based heterogeneous ensemble.
- Binary and multiclass evaluation under one reproducible framework.
- Cross-dataset and multi-location external validation.
- Noise, duration and confidence-based robustness analysis.
- Accuracy-complexity trade-off with feature count and inference time.

Minimum result package for Q1/IEEE paper:

- Dataset summary and class-distribution table.
- Individual model vs ensemble comparison.
- Feature ablation study.
- Baseline vs search-optimized comparison.
- Binary confusion matrix, ROC and PR curves.
- Multiclass confusion matrix and macro-F1.
- External CirCor validation.
- Complexity table.
- Noise and duration robustness results.
- Limitations and no-clinical-overclaim statement.

17. Thesis Result Requirements

Thesis Chapter	Required Content
Chapter 1	Problem, objectives, scope, expected outcomes and screening-only boundary
Chapter 2	PCG signal processing, heart-sound classification, ensemble ML, feature selection, optimization and external validation literature
Chapter 3	Datasets, preprocessing, 138-feature pipeline, PV-MEPCG architecture, search algorithm and equations
Chapter 4	Implementation, code modules, experiments, dashboard, model storage and report generation
Chapter 5	Binary, multiclass, ablation, optimization, external validation, robustness, complexity and failure analysis
Chapter 6	Objective-wise achievement, limitations, conclusion and future clinical validation

18. Developer Task Breakdown and Project Structure

Task	Developer Action
Task 1	Download and verify all three dataset families.
Task 2	Create master metadata file and dataset inventory.
Task 3	Check file integrity, duplicates, sampling rates and labels.
Task 4	Implement mono conversion and 2,000 Hz resampling.
Task 5	Implement Butterworth filtering and z-score normalization.
Task 6	Extract and validate approximately 138 features.
Task 7	Implement fold-safe scaling and feature selection.
Task 8	Train all baseline models.
Task 9	Train and calibrate the proposed soft-voting ensemble.
Task 10	Implement RandomizedSearchCV and one advanced search method.
Task 11	Implement PASCAL Dataset A and B multiclass experiments.
Task 12	Implement CirCor subject-wise external validation.
Task 13	Generate all mandatory tables and graphs.
Task 14	Build dashboard modules.
Task 15	Export report-ready figures, tables and sample reports.
Task 16	Package code, saved models, requirements.txt, README and reproducibility instructions.

Recommended project folder structure:

```
project_root/  
data/  
  raw/  
    physionet2016/  
    pascal_a/  
    pascal_b/  
    circor2022/  
  processed/  
    metadata_master.csv  
notebooks/  
  01_dataset_audit.ipynb  
  02_preprocessing.ipynb  
  03_feature_extraction.ipynb  
  04_binary_models.ipynb  
  05_multiclass_models.ipynb  
  06_search_optimization.ipynb  
  07_external_validation.ipynb  
  08_result_generation.ipynb  
src/  
  data_loader/  
  preprocessing/  
  feature_extraction/  
  feature_selection/  
  models/  
  ensemble/
```

optimization/
evaluation/
explainability/
dashboard/
reporting/
models_saved/
outputs/
tables/
graphs/
confusion_matrices/
reports/
logs/
app.py
requirements.txt
README.md

19. Timeline, Quality Checks and Final Deliverables

Phase	Work	Estimated Time
Phase 1	Dataset download, audit and metadata creation	4-6 days
Phase 2	Preprocessing and feature extraction	5-7 days
Phase 3	Baseline binary models	5-7 days
Phase 4	Ensemble and search optimization	7-10 days
Phase 5	PASCAL multiclass experiments	5-7 days
Phase 6	CirCor external validation	7-10 days
Phase 7	Dashboard and report export	7-10 days
Phase 8	Q1/thesis result packaging	5-7 days
QA Area		Required Check
Dataset QA	File integrity, duplicates, label mapping, class counts and sampling rates	
Split QA	No patient leakage; subject-wise grouping where available	
Feature QA	All features finite, reproducible and extracted identically across folds	
Model QA	Fixed random seed, fold-safe scaling and proper calibration	
Metric QA	Accuracy alone is not sufficient; report sensitivity, specificity, F1, AUC and macro metrics	
Search QA	Search only inside training folds; final test data untouched	
Dashboard QA	Displayed outputs must exactly match backend calculations	
Clinical Safety QA	No diagnosis, treatment, prescription or doctor-replacement claim	
Reproducibility QA	README, environment, seed, configuration and saved outputs included	
Deliverable		Description
Code	Full Python source with requirements.txt and README	

Datasets	Verified folder structure and master metadata
Models	Saved baseline, tuned and ensemble models
Results	All mandatory tables, graphs, confusion matrices and metrics
Optimization	Search logs, best parameters and selected features
External Validation	CirCor and cross-dataset results
Dashboard	Working Streamlit/NiceGUI/Flask interface
Reports	Sample-level and experiment-level exportable reports
Paper Assets	Publication-ready tables, figures and captions
Thesis Assets	Objective mapping, methodology, results and limitation material

Final Developer Notes

- Use PV-MEPCG as the proposed framework name unless the client approves another final name.
- The minimum mandatory predictive models are Logistic Regression, SVM-RBF, Random Forest, Gradient Boosting and Soft Voting Ensemble.
- Use RandomizedSearchCV as the minimum search method and add Bayesian Optimization, GA or PSO if feasible.
- Keep binary, PASCAL multiclass and CirCor murmur/outcome experiments separate.
- Do not merge incompatible class labels.
- Do not report perfect accuracy unless independently verified.
- Final model selection must prioritize balanced performance and sensitivity.
- All tables and graphs listed as mandatory must be generated from code, not manually invented.
- AI-assisted coding may be used, but every generated component must be tested against real data and reproducible outputs.
- The final system must remain an academic research prototype and screening/decision-support framework.